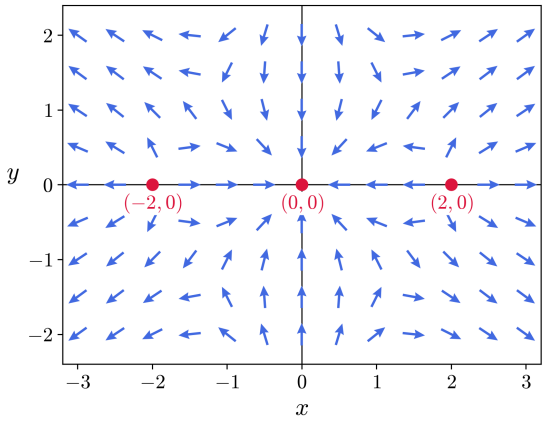
Math 1200 Lecture 18

Nov 5 2026

Topics: 16.5-16.6: Divergence, parametric surfaces Name:_____
Collaborators:_____

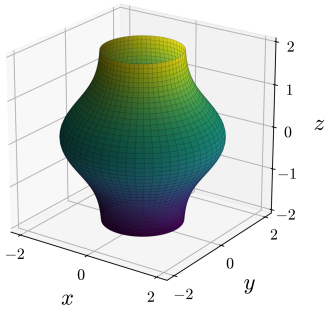
Group work 1 The vector field $\vec{F} = \langle x^3 - 4x + 2xy^2,\ 2y(x^2 - 2)\rangle$ is plotted below. Guess whether $\operatorname{div}(\vec{F})$ is positive or negative at each of the labeled points $(-2, 0)$, $(0, 0)$, $(2, 0)$. Verify your guesses by calculating.



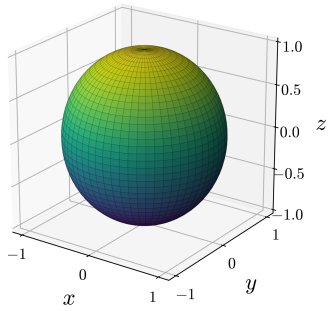
Group work 2 Verify the equation $\nabla \cdot (\nabla \times \vec{F}) = 0$ for the vector field $\vec{F} = \langle -yxz, x^2 + yz, xy\rangle$.

Group work 3 Match the parametric surfaces I–VI with their plots A–F.

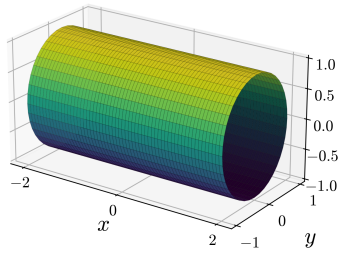
A



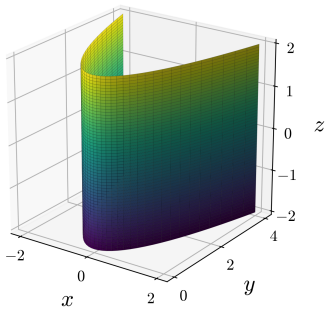
B



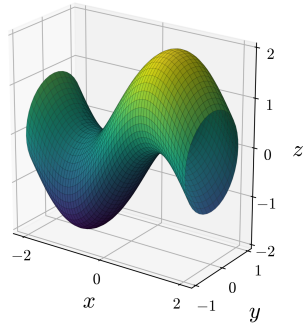
C



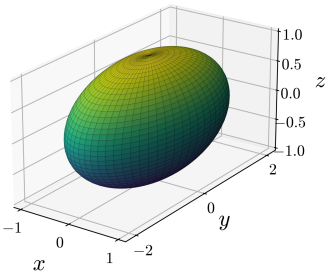
D



E



F



I. $\vec{r}(u,v) = \langle \sin u \cos v,\ \sin u \sin v,\ \cos u \rangle$ $\quad 0 \le u \le \pi,\ 0 \le v \le 2\pi$ _____

II. $\vec{r}(u,v) = \langle u,\ u^2,\ v \rangle$ $\quad -2 \le u \le 2,\ -2 \le v \le 2$ _____

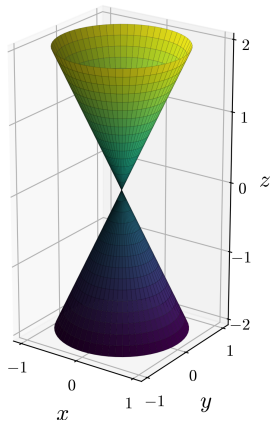
III. $\vec{r}(u,v) = \langle \sin u \cos v,\ 2 \sin u \sin v,\ \cos u \rangle$ $\quad 0 \le u \le \pi,\ 0 \le v \le 2\pi$ _____

IV. $\vec{r}(u,v) = \langle (e^{-u^2}+1)\cos v,\ (e^{-u^2}+1)\sin v,\ u \rangle$ $\quad -2 \le u \le 2,\ 0 \le v \le 2\pi$ _____

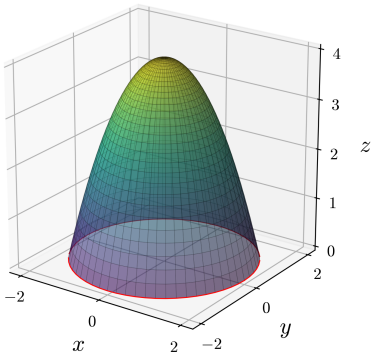
V. $\vec{r}(u,v) = \langle v,\ \cos u,\ \sin(\frac{\pi}{2}v) + \sin u \rangle$ $\quad 0 \le u \le 2\pi,\ -2 \le v \le 2$ _____

VI. $\vec{r}(u,v) = \langle v,\ \cos u,\ \sin u \rangle$ $\quad 0 \le u \le 2\pi,\ -2 \le v \le 2$ _____

Group work 4



a) Parametrize the surface $\vec{r}(u,v)$ which is the cone between $-2 \leq z \leq 2$ of slope 2 (i.e. its intersection with $y = 0$ is two lines of slope ± 2.



b) Parametrize the surface $\vec{r}(u,v)$ which is the part of the graph of $f(x,y) = 2 - x^2 - y^2$ above the xy-plane. **You must get the bounds right, which is tricky!**

Group work 5 (Challenge problem) Parametrize the ellipsoid

$$\frac{x^2}{1} + \frac{y^2}{2} + \frac{z^2}{3} = 1$$